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Document:

Description of the
DL1300
Data logging System

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Introduction

DL1300 is a solution for solving the problem of gathering data from and industrial installation using off-the-shell RS232 and RS485 instrumentation, with a number of important features:

1. No programming required:
Only a XML configuration file is required to set all details about connections, communication lines, mapping physical channels to signal names, setting sampling period, etc.
2. Specific treatment of every class of signals to comply with the following requirements:
 - a. Minimizing the size of the collecting files.
 - b. Collecting enough number of samples for analog signals
 - c. Collect all changes of digital signals
3. Record time per effective sample with resolution better than millisecond, keeping full date description
year-month-day hours:minutes:seconds.milliseconds
4. Tolerant to crashes in the connections:
The logger detects when a given communication line is broken, but this situation does not stop logging through the remaining lines, and automatically continue logging when the line is up again.
5. Software built thinking on the difficulties typically found in the field.

Sampling strategies

Sampling of the devices connected to each line is not synchronous but tailored to the desired sampling rate in each channel.

Digital signals are sampled at high rate but the actual store values correspond only to the changes.

Analog signals sampling strategy is different: the specified sampling rate is used to gather new samples. Actual storage of the sample is also differential: no storage for consecutive samples with the same values, except for the last sample.

Signals are grouped by devices.



Actual Storage of values

The values of the samples (actual value of the signal and time instant of that value) are stored in structured XML format, hence, easy to read ASCII format.

In order to avoid the possibility of losses for big amount of data, these values are stored “per communication line” files. Example: assuming the configuration file contains

```
<Configuracion>
  <linea puerto='/dev/ttyS4' velocidad='38400' nombre='A201' />
  <linea puerto='/dev/ttyS5' velocidad='19200' nombre='CVMK' />
  <linea puerto='/dev/ttyS6' velocidad='38400' nombre='A101' />
  <Dispositivo nombre='53-1-Sup' tipo='6053' linea='A201' direccion='1' />
  <Dispositivo nombre='53-3-Sup' tipo='6053' linea='A201' direccion='3' />
  <Dispositivo nombre='53-4-Sup' tipo='6053' linea='A201' direccion='4' />
  <Dispositivo nombre='17-6-Sup' tipo='6017' linea='A201' direccion='8' />
  <Dispositivo nombre='cvr-15-Sup' tipo='CVR' linea='A201' direccion='15' />
  <Dispositivo nombre='53-10-Inf' tipo='6053' linea='A101' direccion='10' />
  <Dispositivo nombre='53-11-Inf' tipo='6053' linea='A101' direccion='11' />
  <Dispositivo nombre='cvr-14-Inf' tipo='CVR' linea='A101' direccion='14' />
  <Dispositivo nombre='cvmk4c-8' tipo='CVMK4C' linea='CVMK' direccion='8' />
  ...
</Configuracion>
```

Then the generated files will have the names as follows:

2004-06-15_17_18_22.CVMK

2004-06-15_17_18_22.A201

2004-06-15_17_18_22.A101

Etc.

These names include the instant for the first sample included in the file, and the “line” name.

There is a limit in the size of these files that can freely fixed by the user. This limit assures that after surpassing it, the file will be closed and a new file will be opened when required, with its corresponding name. This technique minimizes the possibilities of losing sampled data. I.e., in one of these files is lost then only a limited portion of sampled data is lost.

These files are compressed “on-the-fly” by a low priority process, thus minimizing the required disc size.



Auxiliary programs

DataloggerCommand

This program runs on a Windows PC and connects to the Linux machine running the DL1300 program through sockets and allow the control of the operation of the datalogging process.



With this program you can checked the status of the DL1300 process, check the configuration file in use, order to stop and start datalogging and can stop the program itself.

Note that the DL1300 is executed automatically that the Linux machine is up.

XML to MSAccess transformer

The program reads the (compressed XML) files generated by DL1300 software and create a data base compatible with MSAccess, with a table per signal in those files.

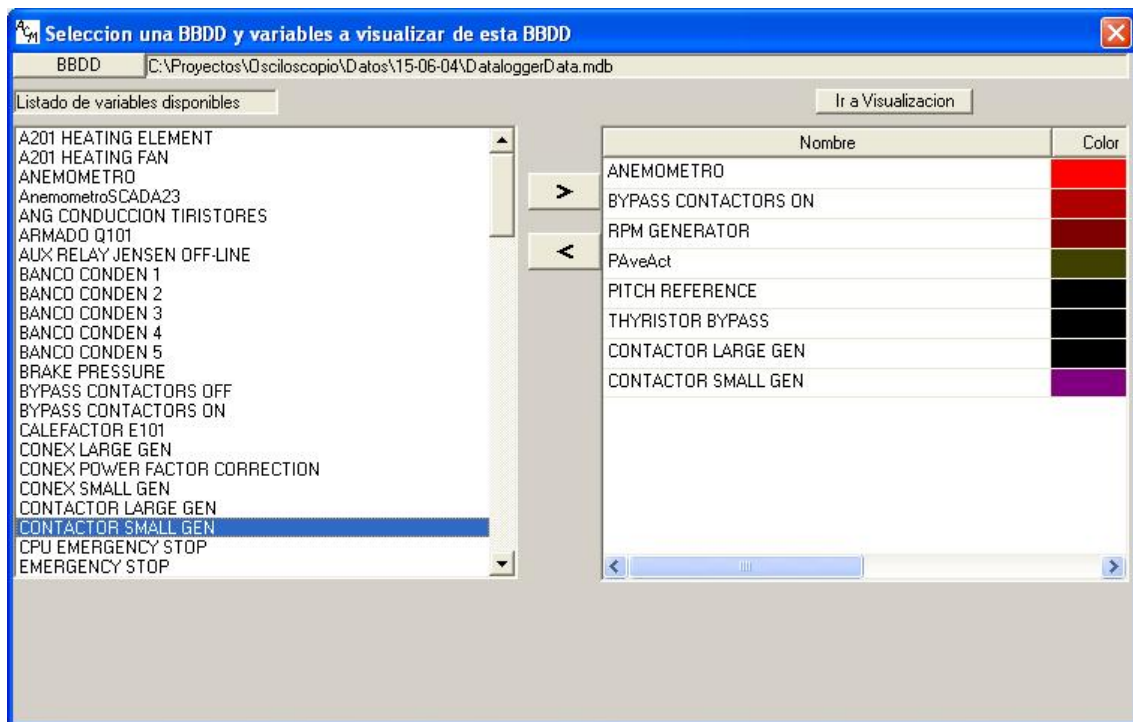


Osciloscopio

This program is designed to ease the presentation of a *big amount of data from big number of signals*.

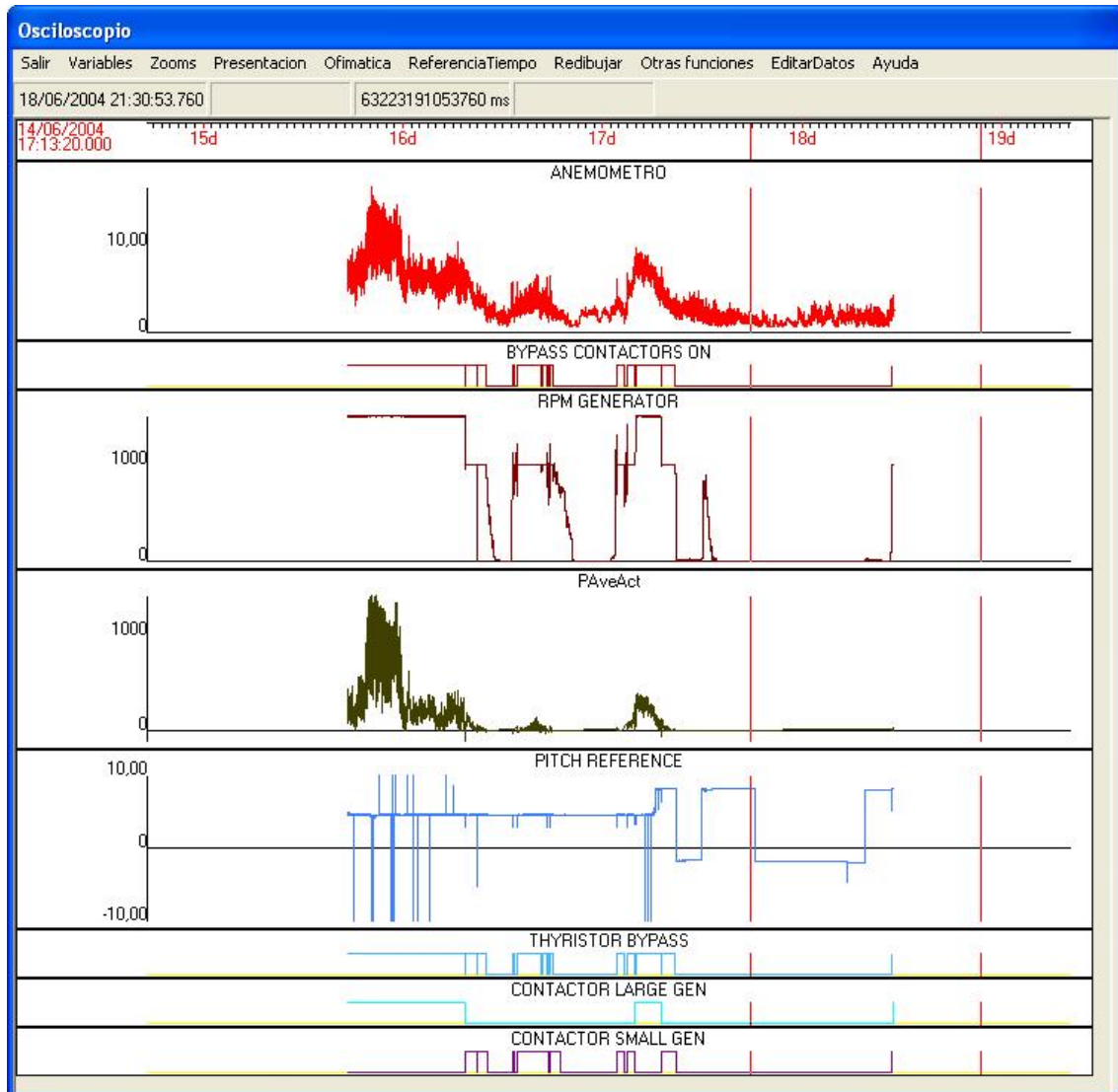
It can also be used to edit sampled data as occasionally errors may appears due to a number of reasons (noise mainly). That task can be tedious and difficult, especially if the number of data is big.

The user may select *any number* of signals to be visualized.





The application finds the time span for the selected signals and makes an adjusted overall view:



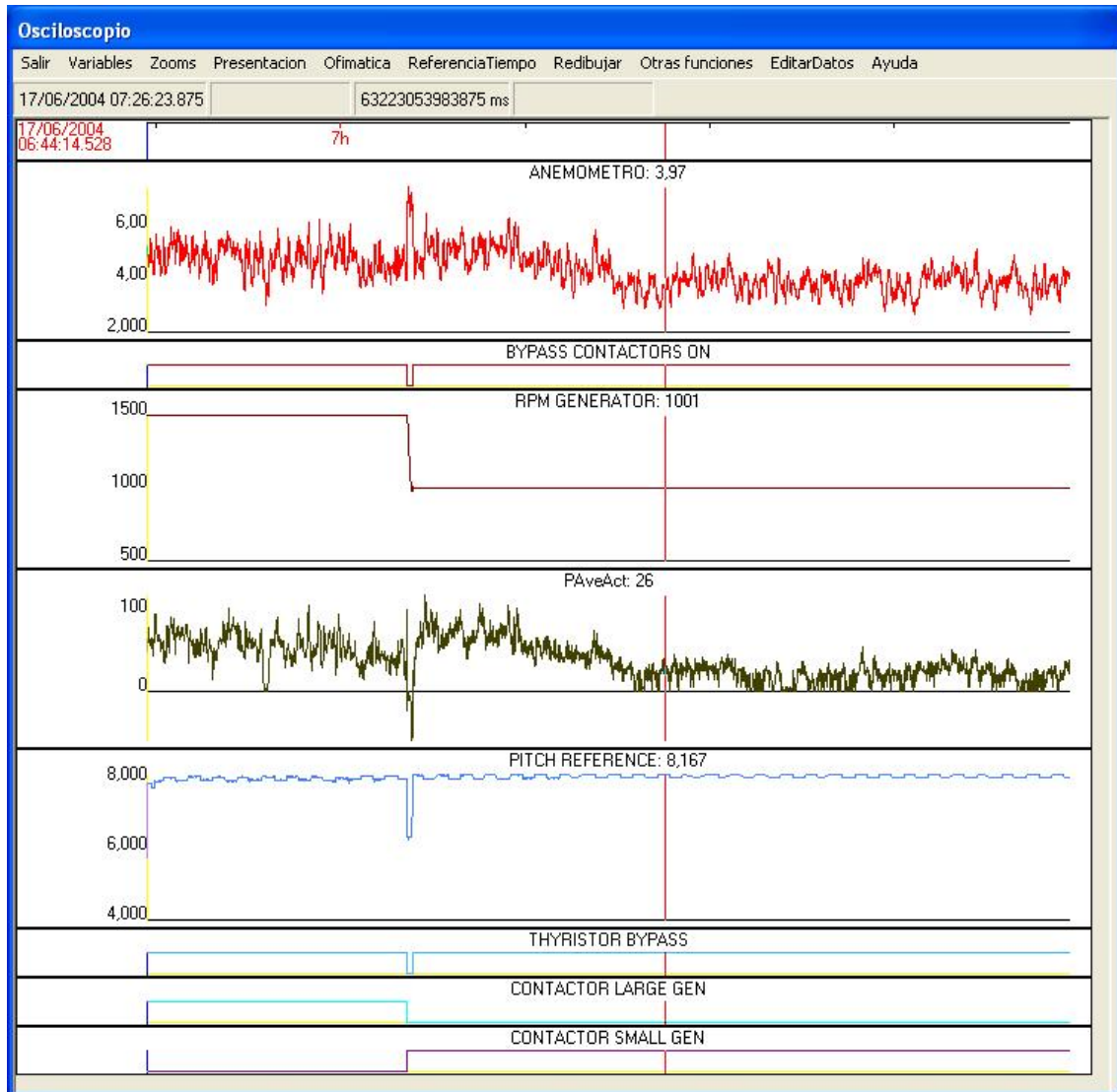
Each signal is presented in its only "channel". The value scale for each channel is self-adjusting, following the typical 1x,2x,5x of a standard oscilloscope. The name of each signal is printed in every channel for easy identification.

Upper channel indicates the time base, adjusting the notchs text accordingly to the most appropriated time span in each moment.



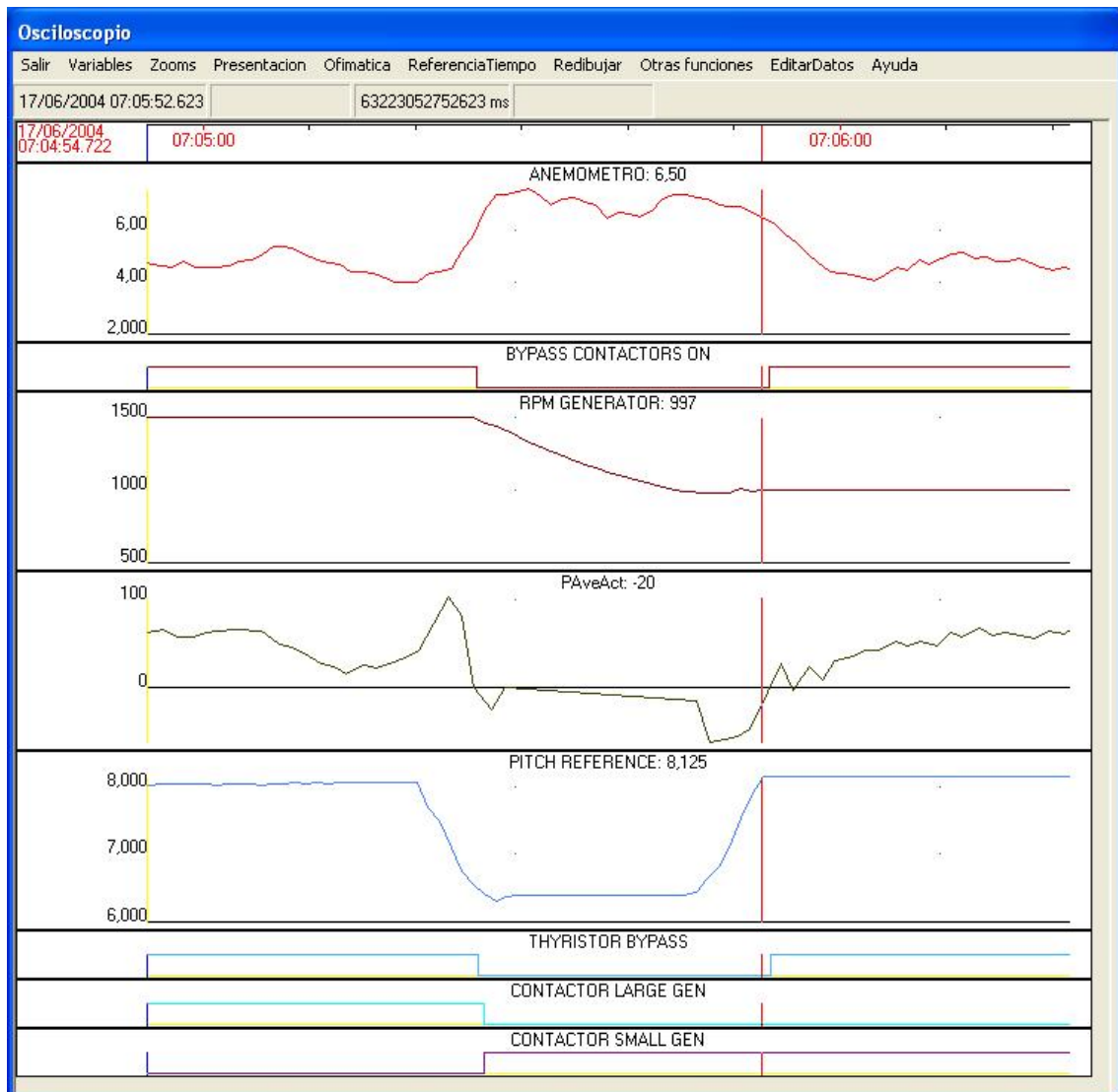
As the mouse moves, a vertical cursor line shows the actual value in each channel, and the corresponding time instant.

The user have several possibilities for making “zooms” (by interactive window definition, by fixed coefficient, panning).



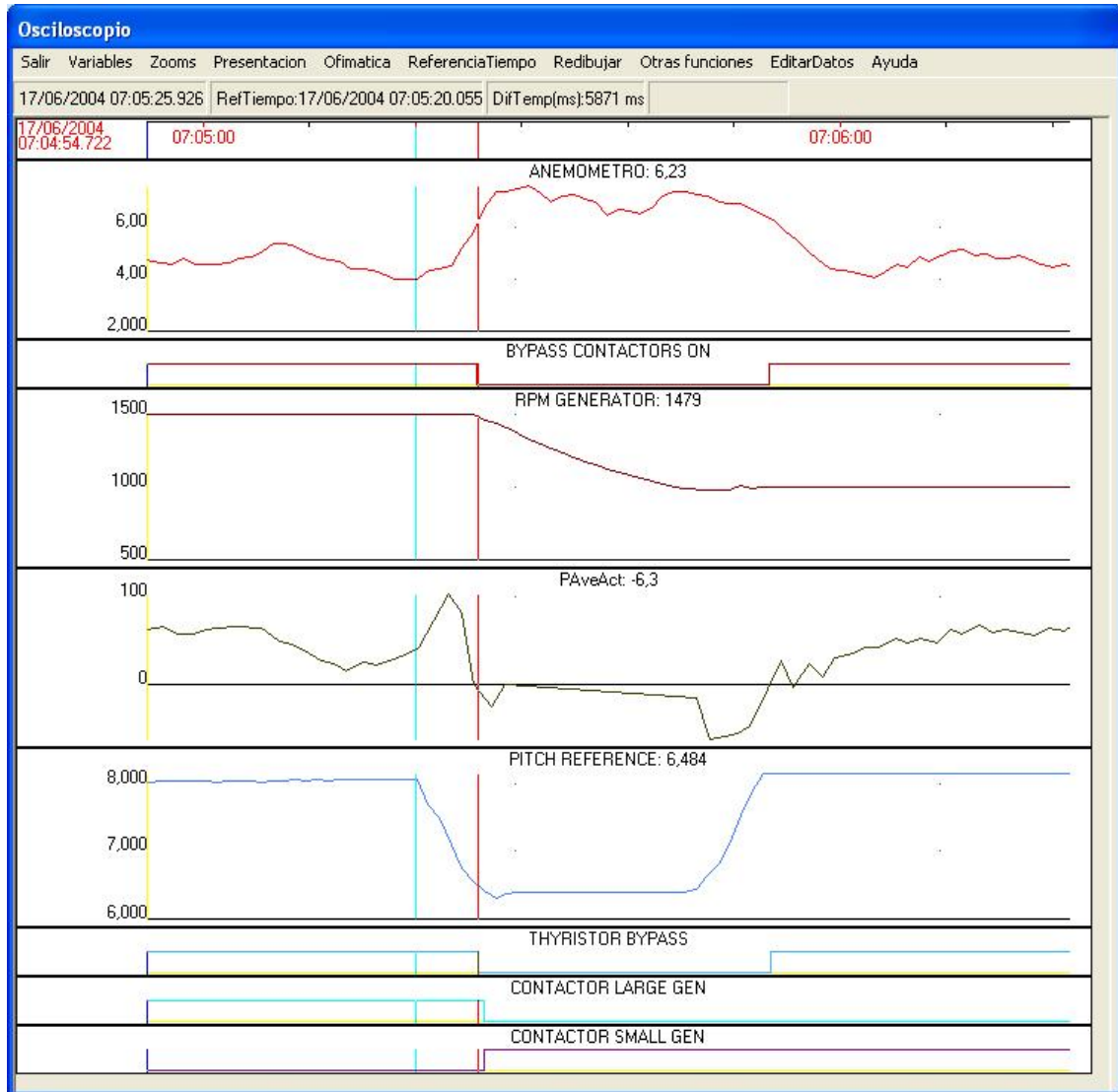


Time windows may have an associated name to jump to it in a fast way.





Time distance can be measured easily defining a “time reference” (blue vertical line in the next figure). After that the time distance is shown as the cursor moves in the “DifTiemp(ms)” box.





Osciloscopio includes a complete rich text editor that “inherits” information from the virtual screen as well as statistical data.

